GATE DA 2027: MACHINE LEARNING MASTER QUESTION BANK

Target: AIR 1 | Comprehensive Chapter-wise Question Bank

Marking: 1 Mark (-0.33) & 2 Marks (-0.66) | NAT/MSQ: No Negatives

CHAPTER 1: SUPERVISED LEARNING – REGRESSION & CLASSIFICATION FOUNDATIONS

Q1. [MCQ | 1 Mark] GATE DA 2024

Consider ordinary least squares (OLS) linear regression $y = Xw + \varepsilon$, where X is an $n \times d$ design matrix with rank d ($n > d$). If all eigenvalues of X^TX are strictly positive and the smallest eigenvalue is denoted by λ_{min}, which of the following statements is true regarding Ridge regression estimate $\hat{w}_{ridge} = (X^TX + \lambda I)^{-1}X^Ty$ with $\lambda > 0$?

(A) The variance of $\hat{w}_{ridge}$ is strictly greater than the variance of $\hat{w}_{OLS}$.

(B) As $\lambda \to \infty$, the bias of $\hat{w}_{ridge}$ approaches zero.

(C) The Euclidean norm $||\hat{w}_{ridge}||_2$ is a strictly monotonically non-increasing function of λ.

(D) $\hat{w}_{ridge}$ is an unbiased estimator for any $\lambda > 0$.

Q2. [NAT | 2 Marks] GATE DA 2024

Consider a binary classification problem with 4 training points in $\mathbb{R}^2$: Positive class (+1): $x_1 = (1, 2)$, $x_2 = (2, 3)$ Negative class (−1): $x_3 = (2, 0)$, $x_4 = (3, 1)$ A hard-margin linear Support Vector Machine (SVM) is trained on this data without a bias term (hyperplane passing through the origin: $w^Tx = 0$). The optimal weight vector is $w = (w_1, w_2)^T$. The value of $w_1 + w_2$ (rounded off to two decimal places) is _____.

Q3. [MSQ | 2 Marks] Expected GATE DA 2027

In standard linear regression solved with Lasso (L1-regularization), which of the following statements are CORRECT?

(A) Lasso yields sparse solutions because the L1-ball has non-differentiable corners along coordinate axes.

(B) When features are highly correlated, Lasso tends to select all of them with equal weights.

(C) Lasso objective function is convex, ensuring any local minimum is a global minimum.

(D) If the regularization parameter λ becomes sufficiently large, all parameter weights w become identically zero.

Q4. [NAT | 2 Marks] Expected GATE DA 2027

Given a simple dataset with 4 points for single-variable linear regression $y = w_1x$: (1, 1.8), (2, 4.1), (3, 5.9), (4, 8.2). Using the closed-form ordinary least squares estimator without intercept, the optimal value of w_1 (rounded off to two decimal places) is _____.

Q5. [MCQ | 1 Mark] GATE DA 2025

In soft-margin Support Vector Machines, slack variables ξ_i are introduced with regularization constant $C > 0$. For a training point (x_i, y_i), which of the following conditions implies that the point is strictly on the wrong side of the separating hyperplane?

(A) $\xi_i = 0$

(B) $0 < \xi_i < 1$

(C) $\xi_i = 1$

(D) $\xi_i > 1$

Q6. [NAT | 2 Marks] Expected GATE DA 2027

Consider a linear regression problem with design matrix $X = [[1, 0], [0, 2]]$ and target vector $y = [2, 4]^T$. If Ridge regression is used with regularization parameter $\lambda = 2$, the L2-regularized cost function is $J(w) = ||Xw - y||_2^2 + \lambda||w||_2^2$. The Euclidean norm of the optimal weight vector $||\hat{w}_{ridge}||_2$ (rounded off to two decimal places) is _____.

Q7. [MSQ | 2 Marks] Expected GATE DA 2027

Let $K(x, z)$ be a kernel function defined on $\mathbb{R}^d \times \mathbb{R}^d$. Which of the following functions are valid Mercer kernels?

(A) $K(x, z) = (x^T z + 2)^3$

(B) $K(x, z) = \exp(-0.5\ ||x - z||_2^2)$

(C) $K(x, z) = x^T z - 5$

(D) $K(x, z) = \exp(x^T z)$

Q8. [NAT | 1 Mark] Expected GATE DA 2027

In a binary classification dataset in $\mathbb{R}^2$, the support vectors for a hard-margin SVM are found to be $s_1 = (1, 1)$ and $s_2 = (2, 0)$ with class label +1, and $s_3 = (3, 3)$ with class label −1. If the optimal weight vector has length $||w||_2 = 2.0$, the geometric margin of the separating hyperplane is _____.

Q9. [MCQ | 2 Marks] Expected GATE DA 2027

In linear regression, let the hat matrix be $H = X(X^T X)^{-1} X^T$. Which of the following statements is FALSE?

(A) H is symmetric ($H = H^T$).

(B) H is idempotent ($H^2 = H$).

(C) The trace of H equals the number of training observations n.

(D) All eigenvalues of H are either 0 or 1.

Q10. [NAT | 2 Marks] Expected GATE DA 2027

Consider a soft-margin SVM dual problem: maximize $\sum_{i=1}^{4} \alpha_i - 0.5 \sum_{i=1}^{4} \sum_{j=1}^{4} \alpha_i \alpha_j y_i y_j (x_i^T x_j)$ subject to $0 \le \alpha_i \le 5$ and $\sum_{i=1}^{4} \alpha_i y_i = 0$. If three optimal dual variables are $\alpha_1 = 1.8$, $\alpha_2 = 5.0$, and $\alpha_3 = 0.0$ with labels $y_1 = 1$, $y_2 = 1$, $y_3 = -1$, $y_4 = -1$, the exact numerical value of α_4 is _____.

Q11. [MSQ | 1 Mark] Expected GATE DA 2027

Which of the following are consequences of severe multicollinearity among predictor variables in Ordinary Least Squares (OLS) regression?

(A) The OLS estimator $\hat{w}$ remains unbiased.

(B) The matrix $X^T X$ becomes ill-conditioned, leading to huge variances in estimated coefficients.

(C) The training R^2 drops dramatically close to 0.

(D) Small perturbations in training data can cause extreme swings in coefficient signs and magnitudes.

Q12. [NAT | 2 Marks] Expected GATE DA 2027

A dataset has 100 observations and 2 orthogonal features such that $X^T X = 100 \times I_2$. The unregularized OLS estimates are $\hat{w}_1 = 3.0$ and $\hat{w}_2 = -1.5$. If Lasso (L1) regularization is applied with penalty $\lambda = 50$ on the objective function $\sum_{i=1}^{n} (y_i - x_i^T w)^2 + \lambda \sum_{j=1}^{2} |w_j|$, the value of the Lasso estimate $\hat{w}_{1,\ Lasso}$ is _____.

Q13. [MCQ | 2 Marks] Expected GATE DA 2027

Consider polynomial regression with degree d fitted on n points. As degree d increases from 1 to n:

(A) Bias increases and Variance decreases.

(B) Bias decreases and Variance increases.

(C) Both Bias and Variance strictly increase.

(D) Both Bias and Variance strictly decrease.

Q14. [NAT | 2 Marks] Expected GATE DA 2027

In a 2D linear SVM with margin hyperplane $w_1x_1 + w_2x_2 + b = 0$, the two parallel support planes are given by $3x_1 - 4x_2 + 8 = 0$ and $3x_1 - 4x_2 - 2 = 0$. The total margin width between the two canonical bounding planes is _____.

Q15. [MSQ | 2 Marks] Expected GATE DA 2027

In the Karush-Kuhn-Tucker (KKT) conditions for support vector machines with primal parameters (w, b, ξ) and dual parameters (α, μ), which of the following statements are mathematically TRUE?

(A) Complementary slackness requires $\alpha_i [y_i(w^T x_i + b) - 1 + \xi_i] = 0$ for all i.

(B) Points strictly outside the margin boundary have $\alpha_i = 0$.

(C) Points that lie strictly inside the margin or are misclassified have $\alpha_i = C$.

(D) The dual optimization problem is concave and possesses strong duality whenever Slater's condition holds.

Solutions & Explanations: Chapter 1 Foundations (Q1 – Q15)

Q195: (A) OPTICS computes a reachability plot that reveals cluster density structure across variable distances, overcoming DBSCAN's single global ε limitation.

CHAPTER 6: DIMENSIONALITY REDUCTION & MANIFOLD LEARNING (Q196 – Q260)

Q196. [MCQ | 1 Mark] GATE DA 2024

Let X be an n × d centered data matrix (each column has zero mean). Let the sample covariance matrix be $S = (1/n) X^T X$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_d \geq 0$ and corresponding orthonormal eigenvectors $u_1, u_2, \ldots, u_d$. Which of the following statements is FALSE?

(A) The first principal component loading vector u_1 maximizes the sample variance of the projected data.

(B) The total variance of the original features equals the sum of all eigenvalues $\sum_{j=1}^{d} \lambda_j$.

(C) The proportion of total variance explained by the first k principal components is $(\sum_{j=1}^{k} \lambda_j) / (\sum_{j=1}^{d} \lambda_j)$.

(D) The principal component scores $z_j = Xu_j$ and $z_k = Xu_k$ are correlated whenever $j \neq k$.

Q197. [NAT | 2 Marks] GATE DA 2024

Consider a 2D dataset with 4 centered points: $x_1 = (1, 1)^T$, $x_2 = (-1, -1)^T$, $x_3 = (2, 2)^T$, $x_4 = (-2, -2)^T$. The sample covariance matrix is $S = (1/4) \sum_{i=1}^{4} x_i x_i^T$. The largest eigenvalue of S is _____.

Q198. [MSQ | 2 Marks] GATE DA 2025

Which of the following statements regarding Principal Component Analysis (PCA) and Singular Value Decomposition (SVD) of a real centered matrix $X \in \mathbb{R}^{n \times d}$ are TRUE?

(A) If $X = U \Sigma V^T$ is the compact SVD of X, then the right singular vectors V are the eigenvectors of X^TX.

(B) The non-zero singular values σ_i of X are related to the eigenvalues λ_i of X^TX by $\sigma_i = \sqrt{(\lambda_i)}$.

(C) PCA projections can be computed directly from SVD without explicitly forming the d × d covariance matrix.

(D) Performing PCA on unstandardized data is always identical to performing PCA on standardized data with unit variance.

Q199. [NAT | 2 Marks] Expected GATE DA 2027

A 3D dataset has empirical covariance matrix S = [[5, 2, 0], [2, 5, 0], [0, 0, 1]]. The percentage of total variance explained by the first two principal components (rounded off to one decimal place) is _____%.

Q200. [MCQ | 2 Marks] GATE DA 2025

In t-SNE (t-Distributed Stochastic Neighbor Embedding), why is a Student-t distribution with 1 degree of freedom (Cauchy distribution) used to model similarities in the low-dimensional embedding space instead of a Gaussian distribution?

(A) To speed up gradient descent computation from $O(N^2)$ to O(N log N).

(B) To alleviate the "crowding problem" by allowing moderately distant points in high dimension to be placed farther apart in low dimension due to heavier tails.

(C) To guarantee that the cost function remains strictly convex.

(D) To ensure that the low-dimensional projection is an exact orthogonal transformation.

Q201. [NAT | 2 Marks] Expected GATE DA 2027

A centered data matrix $X \in \mathbb{R}^{100 \times 5}$ has singular values $\sigma_1 = 20$, $\sigma_2 = 10$, $\sigma_3 = 5$, $\sigma_4 = 2$, $\sigma_5 = 1$. The proportion of total variance captured by the 2-dimensional truncated SVD subspace (rounded off to three decimal places) is _____.

Q202. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following statements regarding Kernel PCA (KPCA) are TRUE?

(A) Kernel PCA performs linear PCA in a high-dimensional reproducing kernel Hilbert space (RKHS) mapped via $\Phi(x)$.

(B) KPCA solves an eigenvalue problem on the n × n Gram matrix K, where $K_{ij} = k(x_i, x_j)$.

(C) Standard KPCA guarantees exact reconstruction of original input space vectors without solving a pre-image problem.

(D) The Gram matrix K must be centered in feature space before performing spectral decomposition.

Q203. [NAT | 2 Marks] Expected GATE DA 2027

Let a 2 × 2 covariance matrix be S = [[3, 1], [1, 3]]. The first principal component eigenvector is $u_1 = [1/\sqrt{2}, 1/\sqrt{2}]^T$. For a data sample $x = (4, 2)^T$, the first principal component score $z_1 = u_1^T x$ (rounded off to two decimal places; take $\sqrt{2} \approx 1.414$) is _____.

Q204. [MCQ | 1 Mark] Expected GATE DA 2027

What is the reconstruction error objective minimized by standard Principal Component Analysis for projecting n points in $\mathbb{R}^d$ onto a k-dimensional linear subspace (k < d)?

(A) Sum of absolute deviations $\sum_{i=1}^{n} ||x_i - \hat{x}_i||_1$

(B) Sum of squared Euclidean reconstruction errors $\sum_{i=1}^{n} ||x_i - \hat{x}_i||_2^2$

(C) Kullback-Leibler divergence between original and projected empirical distributions

(D) Maximum Chebyshev distance $\max_i ||x_i - \hat{x}_i||_\infty$

Q205. [NAT | 2 Marks] Expected GATE DA 2027

In 2D PCA, let $\lambda_1 = 16$ and $\lambda_2 = 4$ be the eigenvalues of the covariance matrix. The ratio of the length of the major axis to the minor axis of the data dispersion ellipse is $\sqrt{(\lambda_1 / \lambda_2)}$. This ratio is _____.

Q206. [MSQ | 1 Mark] Expected GATE DA 2027

Which of the following dimensionality reduction methods are strictly LINEAR techniques?

(A) Classical Principal Component Analysis (PCA)

(B) Multidimensional Scaling (Classical Metric MDS)

(C) t-Distributed Stochastic Neighbor Embedding (t-SNE)

(D) Linear Discriminant Analysis (LDA) projection

Q207. [NAT | 2 Marks] Expected GATE DA 2027

Let X be an n × 3 matrix where features have sample variances $s_1^2 = 10$, $s_2^2 = 6$, and $s_3^2 = 4$. The trace of the sample covariance matrix Tr(S) is _____.

Q208. [MCQ | 2 Marks] Expected GATE DA 2027

In t-SNE, the cost function minimized across all pairs of points is:

(A) Mean Squared Error: $\sum_i \sum_j (p_{ij} - q_{ij})^2$

(B) Kullback-Leibler divergence: $KL(P \,||\, Q) = \sum_i \sum_j p_{j|i} \ln(p_{j|i} / q_{j|i})$

(C) Hinge loss: $\sum_i \sum_j \max(0, 1 - p_{ij} q_{ij})$

(D) Wasserstein distance: $W_1(P, Q)$

Q209. [NAT | 2 Marks] Expected GATE DA 2027

In a dataset with d = 10 continuous variables, PCA is applied to the 10 × 10 sample correlation matrix R. The sum of all 10 eigenvalues of matrix R is _____.

Q210. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following statements about UMAP (Uniform Manifold Approximation and Projection) compared to t-SNE are TRUE?

(A) UMAP is grounded in Riemannian geometry and algebraic topology.

(B) UMAP generally preserves more of the global data structure than standard t-SNE.

(C) UMAP typically runs significantly faster on large datasets than standard t-SNE.

(D) UMAP is a convex optimization problem with a unique global minimum.

Q211. [NAT | 1 Mark] Expected GATE DA 2027

A data matrix X has rank r = 4. The number of strictly positive singular values of X is _____.

Q212. [MCQ | 2 Marks] Expected GATE DA 2027

According to the Eckart-Young-Mirsky theorem, the best rank-k approximation X_k to a matrix X in both Frobenius norm and spectral norm is obtained by:

(A) Setting the last (d − k) diagonal entries of the covariance matrix to zero.

(B) Truncating the SVD expansion $X = \sum_{i=1}^{r} \sigma_i u_i v_i^T$ to the first k terms: $X_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T$.

(C) QR decomposition followed by Gram-Schmidt orthogonalization.

(D) Cholesky factorization with diagonal pivoting.

Q213. [NAT | 2 Marks] Expected GATE DA 2027

Consider matrix A = [[3, 0], [0, −4]]. The Frobenius norm of A, defined as $||A||_F = \sqrt{(\sum_i \sum_j A_{ij}^2)}$, is _____.

Q214. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following conditions necessitate standardizing features (subtracting mean and dividing by standard deviation) prior to fitting PCA?

(A) Features are measured in vastly different physical units (e.g., annual income in dollars vs age in years).

(B) One feature has a numerical variance several orders of magnitude larger than all other features solely due to unit scaling.

(C) The dataset is drawn from an isotropic standard Gaussian distribution N(0, I).

(D) To prevent high-variance variables from dominating the leading principal components arbitrarily.

Q215. [NAT | 2 Marks] Expected GATE DA 2027

In 2D space, a point $x = (3, 4)^T$ is projected onto a 1D subspace spanned by the unit vector $u = (0.6, 0.8)^T$. The Euclidean distance from point x to its orthogonal projection $p = (u^T x)\, u$ is _____.

Q216. [MCQ | 1 Mark] Expected GATE DA 2027

In Linear Discriminant Analysis (LDA) used as a supervised dimensionality reduction method for a C-class problem, what is the maximum number of non-zero discriminant dimensions (eigenvalues) achievable?

(A) C

(B) min(d, C − 1)

(C) d + C

(D) C × (C − 1) / 2

Q217. [NAT | 2 Marks] Expected GATE DA 2027

In PCA, let data matrix X have singular values σ = [10, 6, 2]. If rank-1 approximation X_1 is used, the squared Frobenius error $||X - X_1||_F^2 = \sigma_2^2 + \sigma_3^2$ is _____.

Q218. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following statements about Isomap (Isometric Feature Mapping) are CORRECT?

(A) It constructs a k-nearest neighbor neighborhood graph over the data points.

(B) It approximates geodesic manifold distances using shortest path algorithms (e.g., Dijkstra or Floyd-Warshall).

(C) It applies classical Multidimensional Scaling (MDS) to the resulting geodesic distance matrix.

(D) It assumes the underlying data manifold is non-isometric to a Euclidean space.

Q219. [NAT | 2 Marks] Expected GATE DA 2027

In 2D PCA, the covariance matrix has eigenvalues $\lambda_1 = 8$ and $\lambda_2 = 2$. If a data point has principal component scores $z_1 = 2$ and $z_2 = -1$, the Mahalanobis distance $d_M = \sqrt{((z_1^2 / \lambda_1) + (z_2^2 / \lambda_2))}$ is _____.

Q220. [MCQ | 2 Marks] Expected GATE DA 2027

In PCA centering, if the original data matrix X is not mean-centered before computing the singular value decomposition, what does the first singular vector represent?

(A) The true direction of maximum sample variance.

(B) A vector primarily pointing towards the global mean of the data cloud from the origin.

(C) The eigenvector with the smallest eigenvalue.

(D) An arbitrary null space vector.

Q221. [NAT | 2 Marks] Expected GATE DA 2027

A symmetric positive semi-definite kernel Gram matrix K for 3 points is given by K = [[2, 1, 0], [1, 2, 1], [0, 1, 2]]. The sum of the eigenvalues of K is _____.

Q222. [MSQ | 1 Mark] Expected GATE DA 2027

Which of the following hyperparameter settings significantly impact the low-dimensional embedding produced by t-SNE?

(A) Perplexity (which balances attention between local and global aspects)

(B) Learning rate (step size in gradient descent)

(C) Number of iterations

(D) Kernel ridge penalty λ

Q223. [NAT | 2 Marks] Expected GATE DA 2027

In Locally Linear Embedding (LLE), a data point x_i is reconstructed from its k neighbors as $\hat{x}_i = \sum_{j \in N(i)} W_{ij} x_j$. The weights are constrained to sum to 1: $\sum_j W_{ij} = 1$. If point $x_1 = 4$ is reconstructed from two neighbors $x_2 = 2$ and $x_3 = 8$ with exact match $x_1 = W_{12} x_2 + W_{13} x_3$, the value of weight W_{12} (rounded off to two decimal places) is _____.

Q224. [MCQ | 1 Mark] Expected GATE DA 2027

Why is t-SNE generally unsuitable as a preprocessing dimensionality reduction step for downstream supervised machine learning models on new test data?

(A) It is purely linear and cannot capture non-linear relationships.

(B) It does not learn an explicit parametric mapping function $f: \mathbb{R}^d \to \mathbb{R}^k$ to project unseen out-of-sample test points.

(C) It requires labeled target data to compute pairwise similarities.

(D) It only works on 1-dimensional projections.

Q225. [NAT | 2 Marks] Expected GATE DA 2027

A 5-dimensional data matrix has covariance matrix eigenvalues: $\lambda_1 = 50$, $\lambda_2 = 30$, $\lambda_3 = 10$, $\lambda_4 = 6$, $\lambda_5 = 4$. The minimum number of principal components required to explain at least 85% of total variance is _____.

Q226. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following statements about PCA loading vectors (eigenvectors) u_j and score vectors z_j are mathematically ACCURATE?

(A) The loading vectors are mutually orthogonal: $u_j^T u_k = 0$ for $j \neq k$.

(B) The sample covariance between score vectors z_j and z_k is 0 for $j \neq k$.

(C) The variance of score vector z_j equals the j-th eigenvalue λ_j.

(D) The signs of the eigenvectors are uniquely determined and cannot be flipped.

Q227. [NAT | 2 Marks] Expected GATE DA 2027

In SVD decomposition of matrix $X = U \Sigma V^T$, suppose Σ is a 3×3 diagonal matrix with entries 12, 5, 0. The spectral norm (2-norm) of X, defined as $||X||_2 = \max_{v \neq 0} (||Xv||_2 / ||v||_2)$, is _____.

Q228. [MCQ | 2 Marks] Expected GATE DA 2027

In Kernel PCA, centering the Gram matrix $K \in \mathbb{R}^{n \times n}$ is achieved by which of the following matrix transformations, where 1_n is an $n \times n$ matrix with all elements equal to $1/n$?

(A) $\tilde{K} = K - 1_n K - K 1_n + 1_n K 1_n$

(B) $\tilde{K} = K - \text{diag}(K)$

(C) $\tilde{K} = K^{-1} (I - 1_n)$

(D) $\tilde{K} = 1_n K 1_n$

Q229. [NAT | 1 Mark] Expected GATE DA 2027

In PCA on a dataset with d = 8 variables, the Kaiser criterion suggests retaining all principal components whose eigenvalues are strictly greater than:

(A) 0

(B) 1 (for standardized correlation matrix)

(C) 8

(D) 0.5

Q230. [NAT | 2 Marks] Expected GATE DA 2027

Consider 3 centered points in $\mathbb{R}^2$: $(-2, 0)$, $(0, 0)$, $(2, 0)$. All points lie on the x-axis. The second principal component eigenvalue λ_2 is _____.

Q231. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following geometric interpretations of PCA are mathematically EQUIVALENT for centered data?

(A) Finding orthogonal projection directions that maximize the variance of the projected data points.

(B) Finding a linear subspace of dimension k that minimizes the sum of squared reconstruction errors to original data points.

(C) Finding the orthogonal axes of the best-fitting hyper-ellipsoid to the data cloud.

(D) Finding projection directions that maximize the mutual information between input features and target labels.

Q232. [NAT | 2 Marks] Expected GATE DA 2027

A symmetric covariance matrix S has eigenvalues $\lambda_1 = 9$ and $\lambda_2 = 1$. The condition number of matrix S with respect to the 2-norm, $\kappa(S) = \lambda_{max} / \lambda_{min}$, is _____.

Q233. [MCQ | 1 Mark] Expected GATE DA 2027

What is the computational complexity of performing exact PCA on an n × d matrix (n > d) via eigenvalue decomposition of the sample covariance matrix X^TX?

(A) $O(n\,d^2 + d^3)$

(B) $O(n + d)$

(C) $O(2^d)$

(D) $O(n^3)$

Q234. [NAT | 2 Marks] Expected GATE DA 2027

In 2D, let the data covariance matrix be S = [[4, 0], [0, 1]]. The normalized first eigenvector is $u_1 = (1, 0)^T$. For point $x = (3, 2)^T$, the reconstruction error $||x - (u_1{}^T x)\,u_1||^2$ is _____.

Q235. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following statements about autoencoders as non-linear generalizations of PCA are TRUE?

(A) An autoencoder with non-linear activation functions in hidden layers can learn curved, non-linear manifolds.

(B) An autoencoder with linear activations and squared error loss spans the same principal subspace as standard PCA.

(C) The weights of a linear autoencoder are guaranteed to be orthogonal eigenvectors of X^TX.

(D) Non-linear autoencoders can be prone to learning identity functions if capacity is unconstrained.

Q236. [NAT | 2 Marks] Expected GATE DA 2027

A centered matrix X has SVD $X = U \Sigma V^T$. If $\sigma_1 = 8$ and sample size n = 65, the sample variance along the first principal component $s_1{}^2 = \sigma_1{}^2 / (n - 1)$ is _____.

Q237. [MCQ | 1 Mark] Expected GATE DA 2027

In classical Multidimensional Scaling (MDS), given an n × n squared Euclidean distance matrix $D^{(2)}$, the centering matrix $H = I - (1/n)\,1\,1^T$ is used to compute the inner product Gram matrix B as:

(A) $B = -(1/2)\,H\,D^{(2)}\,H$

(B) $B = H\,D^{(2)}\,H$

(C) $B = (D^{(2)})^{-1}$

(D) $B = H + D^{(2)}$

Q238. [NAT | 2 Marks] Expected GATE DA 2027

Consider 3 variables where the correlation matrix is R = [[1, 0.8, 0], [0.8, 1, 0], [0, 0, 1]]. The trace of R is _____.

Q239. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following properties hold for Singular Value Decomposition (SVD) of any real m × n matrix A?

(A) All singular values σ_i are non-negative real numbers.

(B) The left singular vectors U form an orthonormal basis for $\mathbb{R}^m$.

(C) The right singular vectors V form an orthonormal basis for $\mathbb{R}^n$.

(D) SVD exists only for square invertible matrices.

Q240. [NAT | 2 Marks] Expected GATE DA 2027

Let a 2D dataset have covariance matrix S = [[10, 0], [0, 2]]. A new coordinate system is formed by rotating axes by 45 degrees. The trace of the rotated covariance matrix is _____.

Q241. [MCQ | 2 Marks] Expected GATE DA 2027

In Randomized PCA (Halko et al.), how is computational acceleration achieved on very large n × d matrices?

(A) By projecting the matrix onto a lower-dimensional random Gaussian subspace before computing standard SVD.

(B) By ignoring all off-diagonal covariance entries.

(C) By replacing SVD with a single step of K-Means.

(D) By binarizing all continuous feature values.

Q242. [NAT | 2 Marks] Expected GATE DA 2027

A 3 × 3 matrix has eigenvalues 7, 3, 2. The determinant of this matrix (equal to the product of its eigenvalues) is _____.

Q243. [MSQ | 1 Mark] Expected GATE DA 2027

Which of the following dimensionality reduction methods preserve LOCAL topological neighborhoods rather than global pairwise Euclidean distances?

(A) t-SNE

(B) Locally Linear Embedding (LLE)

(C) Laplacian Eigenmaps

(D) Standard linear PCA

Q244. [NAT | 2 Marks] Expected GATE DA 2027

In 2D space, unit vector $u = (\cos\theta, \sin\theta)^T$ with $\theta = 60°$ ($\cos 60° = 0.5$, $\sin 60° = 0.866$). For point $x = (2, 0)^T$, the scalar projection $u^T x$ is _____.

Q245. [MCQ | 1 Mark] Expected GATE DA 2027

In manifold learning, what is the "Swiss Roll" dataset commonly used to illustrate?

(A) The failure of linear PCA to unroll non-linear 2D manifolds embedded in 3D space without overlapping distant folds.

(B) The inability of neural networks to solve non-convex optimization problems.

(C) The advantage of ridge regression over lasso regression.

(D) The failure of decision trees on continuous data.

Q246. [NAT | 2 Marks] Expected GATE DA 2027

A dataset has 4 features with eigenvalues $\lambda = [12, 5, 2, 1]$. The proportion of variance explained by the smallest principal component (rounded off to three decimal places) is _____.

Q247. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following kernels can be employed in Kernel PCA to capture non-linear feature interactions?

(A) Radial Basis Function (RBF) kernel: $k(x, z) = \exp(-\gamma \|x - z\|^2)$

(B) Polynomial kernel: $k(x, z) = (x^T z + c)^d$

(C) Sigmoid kernel: $k(x, z) = \tanh(\alpha\, x^T z + c)$

(D) Linear kernel: $k(x, z) = x^T z$

Q248. [NAT | 1 Mark] Expected GATE DA 2027

If two features X_1 and X_2 have a perfect linear correlation (Pearson r = +1.0), the rank of their 2 × 2 covariance matrix is _____.

Q249. [MCQ | 2 Marks] Expected GATE DA 2027

In PCA, why are the eigenvectors of the covariance matrix guaranteed to be mutually orthogonal?

(A) Because the sample covariance matrix is real and symmetric, and the Spectral Theorem guarantees orthogonal eigenvectors.

(B) Because all data samples have unit norm.

(C) Because the mean vector is identically zero.

(D) Because the number of rows exceeds the number of columns.

Q248. [NAT | 2 Marks] Expected GATE DA 2027

In 3D PCA, eigenvalues are $\lambda_1 = 15$, $\lambda_2 = 3$, $\lambda_3 = 2$. If the data is projected onto the 2D subspace spanned by the first two eigenvectors, the residual reconstruction error variance (variance lost) is _____.

Q251. [MSQ | 1 Mark] Expected GATE DA 2027

Which of the following statements about Independent Component Analysis (ICA) compared to PCA are TRUE?

(A) PCA seeks uncorrelated orthogonal components, whereas ICA seeks statistically independent non-Gaussian components.

(B) ICA exploits higher-order statistics (kurtosis, negentropy), whereas PCA relies solely on second-order statistics (covariance).

(C) ICA cannot separate multiple Gaussian sources because Gaussian distributions are completely determined by second moments.

(D) PCA order of components is ranked by variance, whereas ICA components have no natural intrinsic ordering.

Q252. [NAT | 2 Marks] Expected GATE DA 2027

A centered data vector in 2D is $x = [2, -2]^T$. The unit eigenvector is $u_1 = [1/\sqrt{2}, -1/\sqrt{2}]^T$. The projection of x onto u_1 is $p = (u_1^T x)\, u_1$. The Euclidean norm $||p||_2$ (rounded off to two decimal places; take $\sqrt{8} \approx 2.83$) is _____.

Q253. [MCQ | 2 Marks] Expected GATE DA 2027

In non-negative matrix factorization (NMF), a non-negative data matrix $V \approx W H$ is decomposed into non-negative factors $W \geq 0$ and $H \geq 0$. What distinct property does NMF have over PCA?

(A) It enforces an additive, parts-based representation without negative cancellations.

(B) It guarantees orthogonal basis vectors.

(C) It is solvable via a unique global closed-form linear algebra equation.

(D) It is invariant to arbitrary rotation of the coordinate system.

Q254. [NAT | 2 Marks] Expected GATE DA 2027

In t-SNE, the pairwise probability in the high-dimensional space is $p_{j|i} = \exp(-||x_i - x_j||^2 / (2\sigma_i^2)) / \sum_{k \neq i} \exp(-||x_i - x_k||^2 / (2\sigma_i^2))$. The perplexity is defined as $\text{Perp}(P_i) = 2^{H(P_i)}$ where $H(P_i)$ is the Shannon entropy in bits. If $H(P_i) = 5.0$ bits, the perplexity is _____.

Q255. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following matrices are always guaranteed to be POSITIVE SEMI-DEFINITE?

(A) Sample covariance matrix $S = (1/n)\, X^T X$ for any real matrix X

(B) Gram kernel matrix K generated by an RBF kernel

(C) Unnormalized graph Laplacian matrix $L = D - W$ for an undirected graph with non-negative edge weights

(D) The hat matrix $H = X(X^T X)^{-1} X^T$

Q256. [NAT | 2 Marks] Expected GATE DA 2027

Consider a 3 × 3 matrix A whose singular values are 5.0, 3.0, and 2.0. The trace of matrix A^TA (equal to the sum of squared singular values) is _____.

Q257. [MCQ | 1 Mark] Expected GATE DA 2027

In PCA scree plots, what visual feature is typically inspected to determine the cutoff number of principal components to retain?

(A) The "elbow" or leveling-off point where eigenvalues cease to drop substantially.

(B) The maximum vertical peak.

(C) The point where the curve intersects the y-axis.

(D) The point of maximum negative derivative.

Q258. [NAT | 2 Marks] Expected GATE DA 2027

In 2D space, two features have variance $\text{Var}(X_1) = 4$ and $\text{Var}(X_2) = 9$, with $\text{Cov}(X_1, X_2) = 0$. The first principal component eigenvalue λ_1 is _____.

Q259. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following statements about PCA vs Fisher's Linear Discriminant Analysis (LDA) are TRUE?

(A) PCA is an unsupervised technique, whereas LDA is a supervised technique.

(B) PCA finds directions of maximal overall variance regardless of class membership.

(C) LDA finds directions that maximize between-class scatter relative to within-class scatter.

(D) PCA can produce at most C − 1 projection axes for a C-class dataset.

Q260. [NAT | 2 Marks] Expected GATE DA 2027

In a 2D PCA setting, the total variance is 100. If the first principal component accounts for 75% of the total variance, the eigenvalue of the second principal component λ_2 is _____.

Solutions & Explanations: Chapter 6 (Q196 – Q260)

Q196: (D) Principal component scores are uncorrelated by construction: $\text{Cov}(z_j, z_k) = u_j^T S u_k = u_j^T (\lambda_k u_k) = \lambda_k (u_j^T u_k) = 0$ for all $j \neq k$.

Q197: 5.00 Points: (1,1), (−1,−1), (2,2), (−2,−2). All points lie on line $y = x$. $x_1x_1^T = [[1, 1], [1, 1]]$, $x_2x_2^T = [[1, 1], [1, 1]]$, $x_3x_3^T = [[4, 4], [4, 4]]$, $x_4x_4^T = [[4, 4], [4, 4]]$. Sum = [[10, 10], [10, 10]]. Covariance $S = (1/4)[[10, 10], [10, 10]] = [[2.5, 2.5], [2.5, 2.5]]$. Eigenvalues: $\lambda_1 = \text{Tr}(S) = 2.5 + 2.5 = 5.00$, and $\lambda_2 = 0$.

Q198: (A, B, C) SVD of $X = U \Sigma V^T$ gives $X^TX = V \Sigma^2 V^T$, confirming V are eigenvectors with $\lambda_i = \sigma_i^2$. (D) is false because PCA is not scale-invariant; standardizing variables completely alters the eigenvectors and variance shares.

Q199: 90.9% $S = [[5, 2, 0], [2, 5, 0], [0, 0, 1]]$. Eigenvalues: For block $[[5, 2], [2, 5]]$, $(5-\lambda)^2 - 4 = 0 \Rightarrow 5 - \lambda = \pm 2 \Rightarrow \lambda_1 = 7, \lambda_2 = 3$. Third eigenvalue is $\lambda_3 = 1$. Total variance = 7 + 3 + 1 = 11. Variance of first two = (7 + 3) / 11 = 10 / 11 = 90.909% ≈ 90.9%.

Q200: (B) In high dimensions, the volume of a sphere grows exponentially, creating a severe shortage of space in 2D (crowding problem). The heavy tails of the Student-t distribution $q_{ij} \propto (1 + ||y_i - y_j||^2)^{-1}$ allow moderate dissimilarities to be mapped to much larger low-dimensional distances.

Q201: 0.943 Eigenvalues are proportional to σ_i^2: $\sigma_1^2 = 400$, $\sigma_2^2 = 100$, $\sigma_3^2 = 25$, $\sigma_4^2 = 4$, $\sigma_5^2 = 1$. Sum = 400 + 100 + 25 + 4 + 1 = 530. First two components = (400 + 100) / 530 = 500 / 530 ≈ 0.94339 ≈ 0.943.

Q202: (A, B, D) KPCA performs linear PCA in RKHS via Gram matrix decomposition. The mapping is implicit, so reconstructing pre-images in input space is non-trivial and generally ill-posed (C is false).

Q203: 4.24 $z_1 = (1/\sqrt{2})(4) + (1/\sqrt{2})(2) = 6 / \sqrt{2} = 3\sqrt{2} \approx 3 \times 1.414 = 4.242 \approx 4.24$.

Q204: (B) PCA minimizes the sum of squared Euclidean projection distances (reconstruction error) between data points and their subspace projections.

Q205: 2.00 Ratio of semi-axes = $\sqrt{(\lambda_1 / \lambda_2)} = \sqrt{(16 / 4)} = \sqrt{4} = 2.00$.

Q206: (A, B, D) PCA, classical MDS, and LDA compute linear orthogonal or projective transformations. t-SNE is a non-linear manifold learning technique.

Q207: 20 Tr(S) is the sum of the sample variances along the diagonal: 10 + 6 + 4 = 20.

Q208: (B) t-SNE minimizes the Kullback-Leibler divergence between conditional high-dimensional joint probabilities P and low-dimensional joint probabilities Q.

Q209: 10 A correlation matrix R has ones along the diagonal ($R_{jj} = 1$). The sum of all eigenvalues equals the trace: Tr(R) = 10 × 1 = 10.

Q210: (A, B, C) UMAP uses fuzzy simplicial sets from topological data analysis; it captures better global continuum structure and is computationally faster. UMAP objective is non-convex (D is false).

Q211: 4 The rank of a matrix equals the number of strictly positive singular values.

Q212: (B) Eckart-Young-Mirsky Theorem establishes that keeping the top k singular values and singular vectors yields the optimal low-rank matrix approximation.

Q213: 5.00 $||A||_F = \sqrt{(3^2 + 0^2 + 0^2 + (-4)^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5.00$.

Q214: (A, B, D) Standardization prevents features with arbitrarily large numeric scales from dominating the covariance matrix. An isotropic Gaussian already has equal unit variance (C is unnecessary).

Q215: 0.00 $u^T x = 0.6(3) + 0.8(4) = 1.8 + 3.2 = 5.0$. Projection $p = 5.0 \times (0.6, 0.8)^T = (3.0, 4.0)^T = x$. Distance = $||x - p|| = 0.00$ (the point lies entirely along vector u).

Q216: (B) The between-class scatter matrix S_B has rank at most C − 1 because it is formed by C class means centered around the global mean.

Q217: 40 Error = $\sigma_2^2 + \sigma_3^2 = 6^2 + 2^2 = 36 + 4 = 40$.

Q218: (A, B, C) Isomap assumes the data manifold is isometric to a Euclidean space and preserves geodesic graph distances via classical MDS.

Q219: 1.00 $d_M = \sqrt{((2^2 / 8) + ((-1)^2 / 2))} = \sqrt{(4/8 + 1/2)} = \sqrt{(0.5 + 0.5)} = \sqrt{1} = 1.00$.

Q220: (B) Without centering, the first singular vector aligns with the direction from the origin to the center of mass of the data cloud, confounding mean position with variance.

Q221: 6 The sum of eigenvalues equals the trace: Tr(K) = 2 + 2 + 2 = 6.

Q222: (A, B, C) Perplexity, learning rate, and iterations directly determine t-SNE optimization dynamics and cluster balance.

Q223: 0.67 $x_1 = W_{12}(2) + W_{13}(8)$ with $W_{13} = 1 - W_{12}$. $4 = 2\,W_{12} + 8(1 - W_{12}) = 8 - 6\,W_{12} \Rightarrow 6\,W_{12} = 4 \Rightarrow W_{12} = 4/6 = 2/3 \approx 0.67$.

Q224: (B) t-SNE optimizes low-dimensional coordinates directly via gradient descent for the training points; it has no parametric projection matrix or mapping function to transform new unseen test inputs.

Q225: 2 Total variance = 50 + 30 + 10 + 6 + 4 = 100. 85% of total variance = 85. With k = 1: 50 (50%). With k = 2: 50 + 30 = 80 (80%). With k = 3: 80 + 10 = 90 (90% ≥ 85%). Minimum components = 3.

Q226: (A, B, C) Eigenvectors have arbitrary sign ambiguity (if u is an eigenvector, −u is also an eigenvector with the same eigenvalue), so (D) is false.

Q227: 12.00 The spectral 2-norm of a matrix equals its largest singular value: $\sigma_1 = 12.00$.

Q228: (A) Standard double centering in feature space: $\tilde{K} = (I - 1_n)\,K\,(I - 1_n) = K - 1_n\,K - K\,1_n + 1_n\,K\,1_n$.

Q229: (B) The Kaiser-Guttman criterion retains components with eigenvalues $\lambda > 1$, representing components that capture more variance than a single standardized original variable.

Q230: 0.00 Since all points have y = 0, there is zero variance in the second orthogonal dimension; thus, $\lambda_2 = 0.00$.

Q231: (A, B, C) Maximum variance projection, minimum reconstruction error, and finding the principal axes of the covariance ellipsoid are mathematically equivalent formulations of PCA. PCA is unsupervised and does not maximize mutual information with labels (D is false).

Q232: 9.00 Condition number $\kappa = 9 / 1 = 9.00$.

Q233: (A) Forming X^TX takes $O(n\,d^2)$ operations, and solving the d × d symmetric eigenvalue problem takes $O(d^3)$, totaling $O(n\,d^2 + d^3)$.

Q234: 4.00 $u_1{}^T x = 1(3) + 0(2) = 3$. Projection $p = (3, 0)^T$. Error $= (3-3)^2 + (2-0)^2 = 0 + 4 = 4.00$.

Q235: (A, B, D) The weights of a linear autoencoder span the same subspace as PCA, but they are not individually constrained to be orthogonal or sorted by eigenvalue unless explicitly enforced (C is false).

Q236: 1.00 $s_1{}^2 = 8^2 / (65 - 1) = 64 / 64 = 1.00$.

Q237: (A) In classical MDS, double centering the squared distance matrix yields $B = -(1/2)\,H\,D^{(2)}\,H = X\,X^T$.

Q238: 3 Trace of any 3 × 3 correlation matrix is 1 + 1 + 1 = 3.

Q239: (A, B, C) SVD exists for all rectangular matrices, whether square or singular (D is false).

Q240: 12 Trace is invariant under orthogonal rotations: $\mathrm{Tr}(R\,S\,R^T) = \mathrm{Tr}(S) = 10 + 2 = 12$.

Q241: (A) Randomized SVD sketches the range of matrix X using a random Gaussian projection matrix, reducing the problem to a small matrix where standard SVD is computed rapidly.

Q242: 42 $\det(A) = 7 \times 3 \times 2 = 42$.

Q243: (A, B, C) t-SNE, LLE, and Laplacian Eigenmaps construct k-NN neighborhood graphs to preserve local geometry. Standard PCA preserves global orthogonal variance.

Q244: 1.00 $u^T x = (0.5)(2) + (0.866)(0) = 1.00$.

Q245: (A) Linear PCA flattens the Swiss Roll by projecting overlapping spiral layers onto each other, whereas non-linear manifold methods (Isomap, LLE) successfully unroll the sheet.

Q246: 0.050 Total variance = 12 + 5 + 2 + 1 = 20. Smallest eigenvalue $\lambda_4 = 1$. Share = 1 / 20 = 0.050.

Q247: (A, B, C, D) All listed kernels satisfy Mercer's theorem and can be used in Kernel PCA (with linear kernel reducing to standard PCA).

Q248: 1 When two features are collinear (r = 1.0), the second eigenvalue is 0, giving a rank of 1.

Q249: (A) By the Spectral Theorem, any real symmetric matrix has real eigenvalues and mutually orthogonal eigenvectors.

Q250: 2.00 Total variance = 15 + 3 + 2 = 20. Retained variance = 15 + 3 = 18. Variance lost = $\lambda_3 = 2.00$.

Q251: (A, B, C, D) All statements accurately capture the fundamental mathematical distinctions between second-order orthogonal PCA and higher-order statistical ICA.

Q252: 2.83 $u_1^T x = (2/\sqrt{2}) + (2/\sqrt{2}) = 4/\sqrt{2} = 2\sqrt{2}$. $p = 2\sqrt{2}\,[1/\sqrt{2}, -1/\sqrt{2}]^T = [2, -2]^T$. Norm = $\sqrt{(2^2 + (-2)^2)} = \sqrt{8} \approx 2.83$.

Q253: (A) Non-negativity constraints forbid negative cancellations, ensuring basis vectors and coefficients represent purely additive component combinations.

Q254: 32 Perp = $2^{5.0} = 32$.

Q255: (A, B, C, D) Covariance matrices, Mercer kernels, undirected graph Laplacians, and orthogonal projection hat matrices are all mathematically positive semi-definite.

Q256: 38 $\mathrm{Tr}(A^T A) = \sum \sigma_i^2 = 5^2 + 3^2 + 2^2 = 25 + 9 + 4 = 38$.

Q257: (A) Scree test plots eigenvalues in descending order to identify the "elbow" where eigenvalues level off into random residual noise.

Q258: 9 Since $\mathrm{Cov}(X_1, X_2) = 0$, the covariance matrix is diagonal: diag(4, 9). Eigenvalues are 9 and 4. The largest is $\lambda_1 = 9$.

Q259: (A, B, C) LDA is bounded by C − 1 dimensions, whereas PCA is bounded by feature dimension d, so (D) is false.

Q260: 25.00 $\lambda_1 = 0.75 \times 100 = 75$. $\lambda_2 = 100 - 75 = 25.00$.

CHAPTER 7: MODEL EVALUATION, VALIDATION & ERROR ANALYSIS (Q261 – Q295)

Q261. [NAT | 2 Marks] GATE DA 2024

A binary classifier is evaluated on an imbalanced test dataset containing 100 positive instances and 900 negative instances. The model correctly identifies 80 positive instances (True Positives = 80) and incorrectly classifies 40 negative instances as positive (False Positives = 40). The F_1-score of the classifier (rounded off to two decimal places) is _____.

Q262. [MCQ | 1 Mark] GATE DA 2024

Which of the following statements is FALSE regarding Receiver Operating Characteristic (ROC) curves and the Area Under the Curve (AUC)?

(A) The ROC curve plots the True Positive Rate (Sensitivity) against the False Positive Rate (1 − Specificity) across varying discrimination thresholds.

(B) An ROC curve is invariant to monotonic increasing transformations of predicted classification scores.

(C) A completely random guess classifier on a balanced binary problem achieves an AUC of 0.5.

(D) If the class distribution changes substantially while keeping the conditional feature distributions fixed, the ROC curve shifts drastically.

Q263. [MSQ | 2 Marks] GATE DA 2025

Which of the following statements about cross-validation techniques are CORRECT?

(A) In Leave-One-Out Cross-Validation (LOOCV) for a dataset of size N, the estimator of generalization error has low bias but typically exhibits high variance.

(B) Stratified K-Fold cross-validation ensures that each fold contains roughly the same proportion of target class labels as the complete dataset.

(C) K-Fold cross-validation training subsets share $(K-2)/(K-1)$ of their training data across folds, which introduces correlation among individual fold error estimates.

(D) Performing feature selection on the entire dataset prior to splitting into cross-validation folds does not cause data leakage.

Q264. [NAT | 2 Marks] GATE DA 2025

A binary classification test set contains 50 positive and 50 negative instances. A model produces the following confusion matrix: TP = 45, FN = 5, FP = 15, TN = 35. The Matthews Correlation Coefficient (MCC), defined as: $MCC = (TP \times TN - FP \times FN) / \sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}$, rounded off to two decimal places, is _____.

Q265. [MCQ | 2 Marks] Expected GATE DA 2027

In statistical learning theory, under squared error loss, the expected prediction error of a model $\hat{f}(x)$ on an unseen test point x with true value $y = f(x) + \varepsilon$ (where $\varepsilon \sim N(0, \sigma^2)$) decomposes into: $\mathbb{E}[(y - \hat{f}(x))^2] = [\text{Bias}(\hat{f}(x))]^2 + \text{Var}(\hat{f}(x)) + \sigma^2$. Which of the following represents the irreducible error?

(A) $[\text{Bias}(\hat{f}(x))]^2$

(B) $\text{Var}(\hat{f}(x))$

(C) σ^2

(D) $[\mathbb{E}[\hat{f}(x)] - f(x)]^2$

Q266. [NAT | 2 Marks] Expected GATE DA 2027

In a ranking task with 4 instances, the ground truth binary labels are y = [1, 0, 1, 0]. A model assigns predicted probability scores s = [0.85, 0.70, 0.60, 0.40]. The Wilcoxon-Mann-Whitney statistic computes AUC as: $AUC = \sum_{i \in Pos} \sum_{j \in Neg} [\, \mathbb{I}(s_i > s_j) + 0.5\, \mathbb{I}(s_i = s_j) \,] / (|Pos| \times |Neg|)$. The AUC for this model is _____.

Q267. [MSQ | 2 Marks] Expected GATE DA 2027

When evaluating classification models under extreme class imbalance (e.g., fraud detection with 0.1% positive class), which of the following metrics are preferred over standard Accuracy and ROC-AUC?

(A) Precision-Recall (PR) Curve and PR-AUC

(B) Matthews Correlation Coefficient (MCC)

(C) Standard classification Accuracy

(D) F_β-score (with $\beta > 1$ to prioritize recall)

Q268. [NAT | 2 Marks] Expected GATE DA 2027

A diagnostic test has Sensitivity = 0.95 and Specificity = 0.90. In a population where disease prevalence is P(Disease) = 0.02, a person tests positive. Using Bayes' theorem, the Positive Predictive Value (Precision) of the test, P(Disease | Positive) (rounded off to three decimal places), is _____.

Q269. [MCQ | 1 Mark] Expected GATE DA 2027

In McNemar's test for comparing two machine learning classifiers A and B evaluated on the exact same test set:

(A) It uses a contingency table of discordant pairs (cases where classifier A is correct and B is wrong, or vice-versa).

(B) It requires training both classifiers on 100 independent bootstrap samples.

(C) It assumes the prediction errors follow a Poisson distribution.

(D) It compares the mean cross-validation accuracy using an independent two-sample t-test.

Q270. [NAT | 2 Marks] Expected GATE DA 2027

A regression model is evaluated on 4 test points with true targets $y = [3.0, 5.0, 7.0, 9.0]$ and predictions $\hat{y} = [2.5, 5.5, 8.0, 8.0]$. The Mean Absolute Percentage Error (MAPE), defined as $(1/n) \sum_{i=1}^{n} |(y_i - \hat{y}_i) / y_i| \times 100\%$, rounded off to one decimal place, is _____%.

Q271. [MSQ | 1 Mark] Expected GATE DA 2027

Which of the following statements about F_β-score: $F_\beta = (1 + \beta^2)$ (Precision × Recall) / (β^2 Precision + Recall) are TRUE?

(A) When $\beta = 1$, it reduces to the harmonic mean of precision and recall.

(B) Setting $\beta = 2$ gives twice as much weight to recall as to precision.

(C) Setting $\beta = 0.5$ gives twice as much weight to precision as to recall.

(D) F_β-score cannot be defined when Precision or Recall is zero.

Q272. [NAT | 2 Marks] Expected GATE DA 2027

For a multi-class problem with 3 classes, the macro-averaged F1 score is the unweighted arithmetic mean of per-class F1 scores. Suppose the F1 scores for Class 1, Class 2, and Class 3 are 0.80, 0.60, and 0.70 respectively. The macro-averaged F1 score is _____.

Q273. [MCQ | 2 Marks] Expected GATE DA 2027

In learning curves plotting training and validation error against training dataset size N, what pattern indicates HIGH VARIANCE (Overfitting)?

(A) Both training and validation errors converge rapidly to a large, unacceptable error plateau.

(B) Training error remains very low, while validation error is significantly higher, leaving a large generalization gap that narrows as N increases.

(C) Validation error drops below training error permanently.

(D) Both curves oscillate randomly without any discernible asymptotic trend.

Q274. [NAT | 2 Marks] Expected GATE DA 2027

A linear regression model is fitted on 100 observations with 4 features. The residual sum of squares is RSS = 200, and the total sum of squares is TSS = 500. The standard coefficient of determination is $R^2 = 1 - (RSS / TSS) = 0.60$. The adjusted R^2, defined as: $R_{adj}^2 = 1 - [(RSS / (n - p - 1)) / (TSS / (n - 1))]$ with n = 100 and p = 4 (rounded off to three decimal places), is _____.

Q275. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following statements regarding Cohen's Kappa coefficient κ are TRUE?

(A) It measures inter-rater agreement for categorical items, accounting for agreement occurring by chance.

(B) κ = 1 indicates perfect agreement between predicted and true labels.

(C) κ ≤ 0 indicates that agreement is no better than chance.

(D) Cohen's Kappa is strictly bounded between 0 and 1.

Q276. [NAT | 1 Mark] Expected GATE DA 2027

In K-Fold cross-validation on a dataset with N = 100 samples, if K = 10, the exact number of test samples evaluated in each fold is _____.

Q277. [MCQ | 2 Marks] Expected GATE DA 2027

What is the purpose of nested cross-validation (double cross-validation)?

(A) To avoid optimistically biased performance estimates when hyperparameter tuning is performed on the same dataset used for evaluation.

(B) To reduce computational runtime from $O(K^2)$ to O(K).

(C) To replace gradient descent with closed-form matrix inversions.

(D) To eliminate the need for an independent test set in infinite sample limits.

Q278. [NAT | 2 Marks] Expected GATE DA 2027

In binary classification, let TP = 40, FP = 10, FN = 20, TN = 130. The Specificity (True Negative Rate = TN / (TN + FP)), rounded off to two decimal places, is _____.

Q279. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following actions directly reduce model BIAS?

(A) Increasing the polynomial degree of regression features

(B) Increasing the maximum depth of a decision tree

(C) Increasing L2 regularization strength λ

(D) Adding informative interaction features to the model

Q280. [NAT | 2 Marks] Expected GATE DA 2027

Consider 5 data points with true binary labels [1, 1, 0, 1, 0]. A model assigns probability scores [0.9, 0.8, 0.7, 0.4, 0.2]. At a decision threshold of τ = 0.5, the precision (TP / (TP + FP)), rounded off to two decimal places, is _____.

Q281. [MCQ | 1 Mark] Expected GATE DA 2027

In Brier score evaluation for probabilistic binary classification: BS = $(1/N) \sum_{i=1}^{N} (f_i - o_i)^2$, where f_i is the forecast probability and $o_i \in \{0, 1\}$. What is the score achieved by a perfect probabilistic model?

(A) 0.0
(B) 0.5
(C) 1.0
(D) −1.0

Q282. [NAT | 2 Marks] Expected GATE DA 2027

A binary classification model produces probability outputs for 2 test samples: Sample 1 (true label 1) gets p_1 = 0.8; Sample 2 (true label 0) gets p_2 = 0.3. The Brier score for these 2 samples is _____.

Q283. [MSQ | 1 Mark] Expected GATE DA 2027

Which of the following evaluation metrics are SCALE-DEPENDENT when assessing regression models?

(A) Root Mean Squared Error (RMSE)
(B) Mean Absolute Error (MAE)
(C) Coefficient of Determination (R^2)
(D) Mean Absolute Percentage Error (MAPE)

Q284. [NAT | 2 Marks] Expected GATE DA 2027

For 5 predictions, the true values are y = [2, 4, 5, 4, 5] and predictions are ŷ = [3, 3, 5, 5, 4]. The Mean Absolute Error (MAE) is _____.

Q285. [MCQ | 2 Marks] Expected GATE DA 2027

In Akaike Information Criterion (AIC = $2k - 2\ln(\hat{L})$) and Bayesian Information Criterion (BIC = $k\ln(n) - 2\ln(\hat{L})$), where k is the number of parameters and n is sample size:

(A) BIC penalizes model complexity more heavily than AIC whenever sample size $n \geq 8$.
(B) AIC penalizes model complexity more heavily than BIC for all sample sizes.
(C) Both criteria favor models with larger numbers of parameters.
(D) AIC is strictly asymptotic while BIC is exact only for $n < 5$.

Q286. [NAT | 2 Marks] Expected GATE DA 2027

A linear regression model has k = 5 parameters. On a dataset of n = 100 samples, the maximized log-likelihood is $\ln(\hat{L}) = -150.0$. Given $\ln(100) \approx 4.605$, the BIC score for this model (rounded off to one decimal place) is _____.

Q287. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following statements about ROC curves are mathematically TRUE?

(A) The point (0, 0) on the ROC curve corresponds to setting the discrimination threshold to +∞ (classifying all instances as negative).
(B) The point (1, 1) corresponds to setting the discrimination threshold to −∞ (classifying all instances as positive).
(C) A classifier that flips all its predictions has an AUC of $1 - AUC_{original}$.
(D) The slope of the tangent line to the ROC curve at any point equals the likelihood ratio at that threshold.

Q288. [NAT | 2 Marks] Expected GATE DA 2027

In an information retrieval setting with 10 documents, 4 documents are relevant. A search engine retrieves 5 documents, of which 3 are relevant. The Recall of the retrieved set (rounded off to two decimal places) is _____.

Q289. [MCQ | 1 Mark] Expected GATE DA 2027

In calibration of classification models, what does a Platt scaling transformation apply to the uncalibrated model scores f(x)?

(A) A scalar logistic sigmoid function P(y = 1 | f(x)) = 1 / (1 + exp(A f(x) + B)).

(B) An isotonic step regression function.

(C) A min-max normalization mapping f(x) to [0, 1].

(D) A Box-Cox power transform.

Q290. [NAT | 2 Marks] Expected GATE DA 2027

In 5-fold cross-validation, the validation accuracy scores across the 5 folds are: [0.82, 0.86, 0.84, 0.88, 0.85]. The sample mean accuracy is 0.85. The sample variance of these 5 validation accuracy estimates (using divisor n − 1 = 4) is _____.

Q291. [MSQ | 1 Mark] Expected GATE DA 2027

Which of the following scenarios cause DATA LEAKAGE during machine learning model development?

(A) Imputing missing values using the mean computed across the entire combined dataset before train-test split.

(B) Standardizing features using the mean and standard deviation of the training set only.

(C) Applying target encoding where class probabilities are computed using target labels from the entire dataset including test folds.

(D) Selecting the top 10 most correlated features with the target using the complete dataset before cross-validation.

Q292. [NAT | 2 Marks] Expected GATE DA 2027

A binary classifier achieves Precision = 0.60 and Recall = 0.80 on a test set. The F_1 score (rounded off to two decimal places) is _____.

Q293. [MCQ | 2 Marks] Expected GATE DA 2027

Consider estimating the test error using bootstrap validation (Efron's 0.632 bootstrap estimator): $Err_{0.632}$ = 0.368 Err_{train} + 0.632 Err_{OOB}. What is the purpose of the 0.632 weighting?

(A) It compensates for the pessimistic bias of the out-of-bag error Err_{OOB} due to smaller effective training set size (~63.2% distinct instances).

(B) It guarantees zero variance in the error estimator.

(C) It bounds the training loss by the VC dimension.

(D) It forces the estimator to equal 5-fold cross-validation.

Q294. [NAT | 2 Marks] Expected GATE DA 2027

In 0.632 bootstrap estimation, suppose a model achieves training error Err_{train} = 0.05 and Out-of-Bag error Err_{OOB} = 0.25. The 0.632 bootstrap estimate of generalization error is _____.

Q295. [MSQ | 2 Marks] Expected GATE DA 2027

Which of the following statements regarding the Vapnik-Chervonenkis (VC) dimension are CORRECT?

(A) The VC dimension of linear classifiers in $\mathbb{R}^d$ with bias is d + 1.

(B) A hypothesis class can shatter a set of N points if it can assign all 2^N possible labelings to those points.

(C) A model with infinite VC dimension can never generalize well under any sample size.

(D) 1D sine classifier f(x) = sign(sin(α x)) has infinite VC dimension despite having only one tunable parameter α.

CHAPTER 8: COMPREHENSIVE MOCK TEST – TARGET AIR 1 (Q296 – Q325)

Q296. [NAT | 2 Marks] Predicted GATE DA 2027

In a Ridge regression setting $y = Xw + \varepsilon$, let X^TX have singular values 20, 10, 5. The effective degrees of freedom of the Ridge fit is defined as $df(\lambda) = Tr(X(X^TX + \lambda I)^{-1}X^T) = \sum_{i=1}^{3} [d_i^2 / (d_i^2 + \lambda)]$ where d_i^2 are eigenvalues of X^TX. If eigenvalues are 20, 10, and 5, and $\lambda = 10$, the effective degrees of freedom df(10) (rounded off to two decimal places) is _____.

Q297. [MCQ | 2 Marks] Predicted GATE DA 2027

Consider a binary classification problem solved using AdaBoost. If at round m the base classifier h_m achieves a weighted training error of exactly $\varepsilon_m = 0.5$, what happens to the instance weights for the next iteration round m + 1?

(A) All instance weights remain completely unchanged.

(B) Misclassified instances receive twice the previous weight.

(C) Correctly classified instances have their weights reset to zero.

(D) All instance weights are inverted: $w_i \leftarrow 1 - w_i$.

Q298. [NAT | 2 Marks] Predicted GATE DA 2027

A soft-margin Support Vector Machine in $\mathbb{R}^2$ uses slack penalty C = 10.0. The optimal dual solution has 3 support vectors with dual multipliers $\alpha_1 = 10.0$, $\alpha_2 = 4.5$, $\alpha_3 = 10.0$. The number of unbounded (margin) support vectors that strictly satisfy $0 < \alpha_i < C$ is _____.

Q299. [MSQ | 2 Marks] Predicted GATE DA 2027

Consider fitting a Gaussian Naive Bayes model where each class conditional distribution is $N(\mu_{c,j}, \sigma_{c,j}^2)$. Which of the following conditions guarantee that the resulting decision boundary between any pair of classes is strictly LINEAR?

(A) Feature variances are identical across all classes for each feature ($\sigma_{c1,j}^2 = \sigma_{c2,j}^2$ for all j).

(B) Class prior probabilities are strictly equal ($P(Y = c_1) = P(Y = c_2)$).

(C) Feature means are identical across classes ($\mu_{c1,j} = \mu_{c2,j}$).

(D) All features are standardized to have zero mean and unit variance globally.

Q300. [NAT | 2 Marks] Predicted GATE DA 2027

In a 2-class logistic regression model, the parameter vector is $w = (w_0, w_1)^T = (-2.0, 0.5)^T$ for model $P(Y = 1 \mid x) = \sigma(w_0 + w_1 x)$. The decision threshold is set to $p^* = 0.5$. The critical value of input feature x at the decision boundary is _____.

Q301. [MCQ | 1 Mark] Predicted GATE DA 2027

Which of the following optimization algorithms is guaranteed to avoid getting trapped in strict saddle points with probability 1 under mild technical conditions (Lee et al., 2016)?

(A) Gradient Descent with random parameter initialization.

(B) Ordinary Least Squares Normal Equation.

(C) Single-step coordinate descent.

(D) Perceptron Learning Algorithm.

Q302. [NAT | 2 Marks] Predicted GATE DA 2027

Consider a 2-layer MLP with 1 hidden layer containing 4 neurons with ReLU activations and a linear scalar output: $\hat{y} = \sum_{j=1}^{4} v_j \max(0, w_j x + b_j)$. For input $x = 2.0$, the hidden neuron parameters are: $(w_1, b_1) = (1.0, -1.0)$, $(w_2, b_2) = (-1.0, 1.0)$, $(w_3, b_3) = (0.5, 0.0)$, $(w_4, b_4) = (2.0, -5.0)$. The output weights are $v = [2.0, 3.0, 4.0, 1.0]^T$. The network output $\hat{y}$ is _____.

Q303. [MSQ | 2 Marks] Predicted GATE DA 2027

Which of the following techniques directly improve training throughput and memory efficiency when training large Transformer models?

(A) FlashAttention (tiling attention computation to reduce HBM memory reads/writes)

(B) Mixed-precision training (FP16/BF16 computation with FP32 master weights)

(C) Gradient checkpointing (activation recomputation during backward pass)

(D) Increasing learning rate by a factor of 1000 without warmup

Q304. [NAT | 2 Marks] Predicted GATE DA 2027

In K-Means clustering in $\mathbb{R}^2$, two cluster centroids are $c_1 = (1, 2)$ and $c_2 = (5, 4)$. The midpoint between the two centroids is $m = (3, 3)$. The equation of the separating hyperplane (perpendicular bisector) is given by $(c_2 - c_1)^T (x - m) = 0$, which can be written as $4x_1 + 2x_2 = C$. The constant C is _____.

Q305. [MCQ | 2 Marks] Predicted GATE DA 2027

In Singular Value Decomposition (SVD) of a matrix $X \in \mathbb{R}^{n \times d}$ with $n > d$, what is the geometric relationship between the column space of X and the left singular vectors U?

(A) The first r columns of U (where r = rank(X)) form an orthonormal basis for the column space of X.

(B) The columns of U span the null space of X^TX.

(C) The right singular vectors V span the column space of X.

(D) U is always a diagonal matrix containing the condition number.

Q306. [NAT | 2 Marks] Predicted GATE DA 2027

A dataset has 4 points with true labels $y = [1, 1, 0, 0]$ and model predicted probabilities $p = [0.9, 0.7, 0.3, 0.1]$. The binary cross-entropy loss is: $L = -(1/4) \sum_{i=1}^{4} [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)]$. Given $\ln(0.9) \approx -0.1054$ and $\ln(0.7) \approx -0.3567$. The value of L (rounded off to three decimal places) is _____.

Q307. [MSQ | 2 Marks] Predicted GATE DA 2027

Which of the following statements about Support Vector Regression (SVR) with an ε-insensitive loss function $L_\varepsilon(y, f(x)) = \max(0, |y - f(x)| - \varepsilon)$ are TRUE?

(A) Any training point that falls strictly inside the ε-tube ($|y_i - f(x_i)| < \varepsilon$) has zero loss.

(B) Points strictly inside the ε-tube have dual Lagrange multipliers $\alpha_i = 0$ and $\alpha_i^* = 0$.

(C) Increasing ε increases the number of support vectors.

(D) Only points on or outside the boundary of the ε-tube contribute to the final regression function.

Q308. [NAT | 1 Mark] Predicted GATE DA 2027

In an AdaBoost ensemble of 5 binary classifiers, the model voting weights are $\alpha = [0.8, 0.6, 0.5, 0.4, 0.3]$. On a test point x, the individual classifier predictions are $h(x) = [+1, -1, +1, -1, +1]$. The sign of the weighted sum $\sum_{m=1}^{5} \alpha_m h_m(x)$ determines the ensemble prediction. The numerical value of the weighted sum is _____.

Q309. [MCQ | 2 Marks] Predicted GATE DA 2027

In Gaussian Mixture Models, what is the consequence of setting the covariance matrices of all K components to be spherical and identical ($\Sigma_k = \sigma^2 I$) with equal mixing weights ($\pi_k = 1/K$)?

(A) The posterior responsibility γ_{ik} depends solely on the Euclidean distance $||x_i - \mu_k||^2$, behaving as a soft-assignment equivalent of K-Means.

(B) The decision boundaries become high-order polynomial surfaces.

(C) The EM algorithm fails to converge.

(D) It becomes mathematically identical to Spectral Clustering.

Q310. [NAT | 2 Marks] Predicted GATE DA 2027

A 3 × 3 covariance matrix S has eigenvalues $\lambda_1 = 12$, $\lambda_2 = 6$, $\lambda_3 = 2$. A low-rank approximation is formed using the first principal component only. The percentage of variance lost (reconstruction error percentage), rounded off to one decimal place, is _____%.

Q311. [MSQ | 2 Marks] Predicted GATE DA 2027

Which of the following conditions lead to an ill-conditioned or non-invertible matrix (X^TX) in Ordinary Least Squares linear regression?

(A) The number of predictor features d is strictly greater than the number of observations n ($d > n$).

(B) Two features are exact linear combinations of each other (perfect multicollinearity).

(C) A feature has zero variance across all training instances (constant feature).

(D) An L2 regularization penalty λI with $\lambda > 0$ is added to X^TX.

Q312. [NAT | 2 Marks] Predicted GATE DA 2027

Consider training a Perceptron with learning rate $\eta = 1.0$. The current weight vector is $w = (1, -1)^T$ and bias $b = 0$. A training point $x = (2, 3)^T$ with true label $y = +1$ is misclassified because $w^Tx + b = 1(2) + (-1)(3) = -1 < 0$. Using the Perceptron update rule $w_{new} = w + y\,x$ and $b_{new} = b + y$, the Euclidean norm of the updated weight vector $||w_{new}||_2$ (rounded off to two decimal places; take $\sqrt{13} \approx 3.61$) is _____.

Q313. [MCQ | 1 Mark] Predicted GATE DA 2027

In t-SNE, what is the role of the "early exaggeration" parameter during the initial phase of optimization?

(A) It multiplies the high-dimensional joint probabilities p_{ij} by a constant factor (> 1) to force natural clusters to form tight, well-separated clumps.

(B) It decays the learning rate to zero within the first 10 steps.

(C) It binarizes all distance matrices.

(D) It projects points onto the first two principal components.

Q314. [NAT | 2 Marks] Predicted GATE DA 2027

In a binary classification model evaluated on 100 samples, the confusion matrix entries are TP = 30, FP = 10, FN = 20, TN = 40. The Balanced Accuracy, defined as (Sensitivity + Specificity) / 2, rounded off to two decimal places, is _____.

Q315. [MSQ | 2 Marks] Predicted GATE DA 2027

Which of the following statements about Convolutional Neural Networks (CNN) vs Vision Transformers (ViT) are TRUE?

(A) CNNs possess strong inductive biases (translation equivariance and locality) built into their architecture.

(B) Vision Transformers have less inductive bias than CNNs and typically require larger training datasets or strong pre-training to outperform CNNs.

(C) ViT splits images into fixed-size patches and treats them as tokens in a sequence.

(D) CNNs cannot be trained with gradient descent.

Q316. [NAT | 2 Marks] Predicted GATE DA 2027

Consider a continuous random variable $X \sim \text{Uniform}(0, \theta)$. A single sample $X_1 = 4.0$ is observed. The maximum likelihood estimator (MLE) of θ is $\hat{\theta}_{MLE} = X_1 = 4.0$. The unbiased minimum variance estimator is $\hat{\theta}_{UMVUE} = ((n + 1)/n)\, X_{(n)}$. For $n = 1$, the value of $\hat{\theta}_{UMVUE}$ is _____.

Q317. [MCQ | 2 Marks] Predicted GATE DA 2027

In Ridge regression, why does the Ridge coefficient path never set any weight w_j to exactly zero for finite $\lambda > 0$ (unlike Lasso)?

(A) Because the L2 ball is smooth and strictly differentiable everywhere, meaning the contours of the quadratic loss touch the constraint sphere tangentially off the coordinate axes.

(B) Because the Ridge penalty is non-convex.

(C) Because Ridge regression requires computing matrix inverses.

(D) Because Ridge regression is equivalent to a Laplace prior.

Q318. [NAT | 2 Marks] Predicted GATE DA 2027

Consider an agglomerative hierarchical clustering with Lance-Williams formula for Complete Linkage: $d(C_{ij}, C_k) = 0.5\, d(C_i, C_k) + 0.5\, d(C_j, C_k) + 0.5\, |d(C_i, C_k) - d(C_j, C_k)|$. If $d(C_1, C_3) = 4.0$ and $d(C_2, C_3) = 7.0$, the merged distance $d(C_{12}, C_3)$ is _____.

Q319. [MSQ | 1 Mark] Predicted GATE DA 2027

Which of the following statements about the Expectation-Maximization (EM) algorithm applied to GMM are TRUE?

(A) The responsibilities γ_{ik} always satisfy $\sum_{k=1}^{K} \gamma_{ik} = 1$ for each data point i.

(B) The estimated mixture weights $\hat{\pi}_k$ in the M-step satisfy $\sum_{k=1}^{K} \hat{\pi}_k = 1$.

(C) The log-likelihood is guaranteed to reach the global maximum under any initialization.

(D) EM is sensitive to the initial values chosen for means, covariances, and mixing proportions.

Q320. [NAT | 2 Marks] Predicted GATE DA 2027

In a 2D dataset, the empirical covariance matrix is S = [[4, 2], [2, 4]]. The sum of the two eigenvalues of S (Tr(S)) is _____.

Q321. [MCQ | 2 Marks] Predicted GATE DA 2027

In deep learning, why does Batch Normalization act as an implicit regularizer during mini-batch training?

(A) Because mini-batch mean and variance fluctuate stochastically across batches, introducing multiplicative and additive noise into hidden representations.

(B) Because it zeros out half of the weights at random like Dropout.

(C) Because it adds an explicit L1 penalty on the gradient vector.

(D) Because it forces the loss function to become strictly convex.

Q322. [NAT | 2 Marks] Predicted GATE DA 2027

A binary classifier outputs predicted probabilities for 3 positive instances: [0.9, 0.6, 0.4] and 2 negative instances: [0.7, 0.2]. In the Wilcoxon-Mann-Whitney AUC calculation, the number of pairs (i, j) where pos score s_i > neg score s_j out of the 3 × 2 = 6 possible pairs is _____.

Q323. [MSQ | 2 Marks] Predicted GATE DA 2027

Which of the following statements regarding the bias-variance trade-off are mathematically ACCURATE?

(A) k-Nearest Neighbors with k = 1 has low bias and high variance.

(B) k-Nearest Neighbors with k = N (total sample size) has high bias and low variance.

(C) In Bagging, increasing the number of trees M decreases the variance without altering individual tree bias.

(D) In Boosting, sequential training of weak learners primarily reduces variance while leaving bias unchanged.

Q324. [NAT | 2 Marks] Predicted GATE DA 2027

In an information retrieval evaluation, the precision at rank k (P@k) measures the proportion of relevant documents among the top k retrieved results. If the retrieved ranking of 5 documents has relevance indicators [Relevant, Non-Relevant, Relevant, Relevant, Non-Relevant], the value of P@3 (rounded off to two decimal places) is _____.

Q325. [MCQ | 2 Marks] Predicted GATE DA 2027

Consider an overparameterized deep neural network that interpolates all training data perfectly (achieving exactly 0 training loss). What phenomenon explains why such interpolating models can achieve state-of-the-art generalization performance on unseen test data without suffering from catastrophic overfitting (Belkin et al., 2019)?

(A) The "Double Descent" phenomenon, where beyond the interpolation threshold, implicit regularization of gradient descent finds minimum-norm interpolators that generalize effectively.

(B) The Central Limit Theorem for finite empirical samples.

(C) The exact equivalence of deep networks to 1-nearest neighbor classifiers.

(D) The collapse of the VC dimension to zero in infinite dimensions.

Solutions & Explanations: Chapters 7 & 8 (Q261 – Q325)

Q261: 0.73 Positives = 100, Negatives = 900. TP = 80 ⇒ FN = 100 − 80 = 20. FP = 40. Precision = TP / (TP + FP) = 80 / (80 + 40) = 80 / 120 = 2/3 ≈ 0.6667. Recall = TP / (TP + FN) = 80 / 100 = 0.80. F1 = 2 × (Precision × Recall) / (Precision + Recall) = 2 × (0.6667 × 0.80) / (0.6667 + 0.80) = 1.0667 / 1.4667 ≈ 0.727 ≈ 0.73.

Q262: (D) ROC curve is formed by True Positive Rate and False Positive Rate, which evaluate each class conditional distribution independently; it is therefore invariant to shifts in class prevalence P(Y).

Q263: (A, B, C) Feature selection on the full dataset before splitting causes information leakage, invalidating validation estimates (D is false).

Q264: 0.60 TP = 45, FN = 5, FP = 15, TN = 35. Numerator = (45 × 35) − (15 × 5) = 1575 − 75 = 1500. Denominator = √((45 + 15)(45 + 5)(35 + 15)(35 + 5)) = √(60 × 50 × 50 × 40) = √(6,000,000) ≈ 2449.49. MCC = 1500 / 2449.49 ≈ 0.612 ≈ 0.61 (Acceptable: 0.60 to 0.62).

Q265: (C) σ^2 represents irreducible error arising from inherent noise in the data-generating process that cannot be eliminated by any model.

Q266: 0.75 Positives: $s_1 = 0.85$, $s_3 = 0.60$. Negatives: $s_2 = 0.70$, $s_4 = 0.40$. Pairs (Pos, Neg): (0.85, 0.70): 0.85 > 0.70 (1.0) (0.85, 0.40): 0.85 > 0.40 (1.0) (0.60, 0.70): 0.60 < 0.70 (0.0) (0.60, 0.40): 0.60 > 0.40 (1.0) Sum = 1 + 1 + 0 + 1 = 3. Total pairs = 2 × 2 = 4. AUC = 3 / 4 = 0.75.

Q267: (A, B, D) Under severe class imbalance, standard accuracy is misleading (a model predicting all negatives can achieve 99.9% accuracy). PR curves, MCC, and F-beta focus heavily on minority positive detection.

Q268: 0.162 Sensitivity $P(+|D) = 0.95$, Specificity $P(-|\bar{D}) = 0.90$ ⇒ False Positive Rate $P(+|\bar{D}) = 0.10$. $P(D) = 0.02$, $P(\bar{D}) = 0.98$. $P(D \mid +) = [P(+|D)\,P(D)] / [P(+|D)\,P(D) + P(+|\bar{D})\,P(\bar{D})] = (0.95 \times 0.02) / [0.95(0.02) + 0.10(0.98)] = 0.019 / (0.019 + 0.098) = 0.019 / 0.117 \approx 0.16239 \approx 0.162$.

Q269: (A) McNemar's test constructs a 2 × 2 contingency table of discordant outcomes between two models evaluated on the same test sample and performs a chi-square test.

Q270: 12.1% Absolute percentage errors: Point 1: |3 − 2.5| / 3 = 0.5 / 3 = 0.1667 (16.67%) Point 2: |5 − 5.5| / 5 = 0.5 / 5 = 0.1000 (10.00%) Point 3: |7 − 8.0| / 7 = 1.0 / 7 = 0.1429 (14.29%) Point 4: |9 − 8.0| / 9 = 1.0 / 9 = 0.1111 (11.11%) Mean = (16.67 + 10.00 + 14.29 + 11.11) / 4 = 52.07 / 4 ≈ 13.0% (using exact fractions: (1/6 + 1/10 + 1/7 + 1/9)/4 = (0.5206)/4 ≈ 13.0%).

Q271: (A, B, C) F_β-score is well-defined and set to 0 when precision and recall are both zero.

Q272: 0.70 Macro-F1 = (0.80 + 0.60 + 0.70) / 3 = 2.10 / 3 = 0.70.

Q273: (B) High variance models overfit the training set (low training error) while lagging substantially on validation data, creating a wide gap.

Q274: 0.583 $R_{adj}^2 = 1 - [\,(200 / (100 - 4 - 1)) / (500 / (100 - 1))\,] = 1 - [\,(200 / 95) / (500 / 99)\,] = 1 - [\,2.10526 / 5.05051\,] = 1 - 0.41684 \approx 0.58316 \approx 0.583$.

Q275: (A, B, C) Cohen's Kappa can take negative values down to −1 (indicating systematic disagreement), so it is not bounded strictly between 0 and 1.

Q276: 10 Fold test size = N / K = 100 / 10 = 10.

Q277: (A) Nested cross-validation uses an inner loop for hyperparameter search and an outer loop for unbiased evaluation of test error.

Q278: 0.93 Specificity = TN / (TN + FP) = 130 / (130 + 10) = 130 / 140 ≈ 0.92857 ≈ 0.93.

Q279: (A, B, D) Increasing model capacity, adding polynomial terms, and feature interactions reduce bias. Increasing regularization increases bias.

Q280: 0.67 At $\tau = 0.5$, predicted positives are instances with $p \geq 0.5$: instances 1, 2, 3 (scores 0.9, 0.8, 0.7). True labels are [1, 1, 0]. TP = 2, FP = 1. Precision = 2 / (2 + 1) = 2/3 ≈ 0.67.

Q281: (A) A perfect forecaster predicts probability 1.0 for all occurrences (1) and 0.0 for non-occurrences (0), yielding a Brier score of 0.0.

Q282: 0.065 Sample 1: $(0.8 - 1)^2 = (-0.2)^2 = 0.04$. Sample 2: $(0.3 - 0)^2 = (0.3)^2 = 0.09$. Brier score = (0.04 + 0.09) / 2 = 0.13 / 2 = 0.065.

Q283: (A, B) RMSE and MAE retain the physical units of the target variable (scale-dependent). R^2 and MAPE are dimensionless proportions.

Q284: 0.80 Deviations: |2−3|=1, |4−3|=1, |5−5|=0, |4−5|=1, |5−4|=1. MAE = (1 + 1 + 0 + 1 + 1) / 5 = 4 / 5 = 0.80.

Q285: (A) Penalty in BIC is $k \ln(n)$; penalty in AIC is $2k$. When $\ln(n) > 2$ (i.e., $n > e^2 \approx 7.39$), BIC imposes a stronger penalty on parameter count.

Q286: 323.0 BIC = $k \ln(n) - 2 \ln(\hat{L}) = 5(4.605) - 2(-150.0) = 23.025 + 300.0 = 323.025 \approx 323.0$.

Q287: (A, B, C, D) All statements are fundamental geometric and mathematical properties of ROC curves.

Q288: 0.75 Recall = (Retrieved Relevant) / (Total Relevant) = 3 / 4 = 0.75.

Q289: (A) Platt scaling fits a logistic regression model to the raw model decision scores to output calibrated probabilities.

Q290: 0.0005 Deviations: $(0.82-0.85)^2 = 0.0009$, $(0.86-0.85)^2 = 0.0001$, $(0.84-0.85)^2 = 0.0001$, $(0.88-0.85)^2 = 0.0009$, $(0.85-0.85)^2 = 0$. Sum = 0.0020. Variance = 0.0020 / 4 = 0.0005.

Q291: (A, C, D) Preprocessing steps involving global target or distribution statistics before splitting lead to test information leakage.

Q292: 0.69 F1 = $2(0.60 \times 0.80) / (0.60 + 0.80) = 0.96 / 1.40 \approx 0.6857 \approx 0.69$.

Q293: (A) Because bootstrap training samples omit ~36.8% of instances, Err_{OOB} is pessimistically biased; the 0.632 rule blends training and OOB errors to balance bias.

Q294: 0.176 Err = 0.368(0.05) + 0.632(0.25) = 0.0184 + 0.1580 = 0.1764 ≈ 0.176.

Q295: (A, B, D) The 1D sine classifier can shatter arbitrary numbers of points, exhibiting infinite VC dimension with 1 parameter. High VC dimension models can still generalize under large sample sizes or algorithmic regularization (C is false).

Q296: 1.50 Eigenvalues d_i^2: 20, 10, 5. With $\lambda = 10$: df = 20/(20+10) + 10/(10+10) + 5/(5+10) = 20/30 + 10/20 + 5/15 = 2/3 + 1/2 + 1/3 = 1 + 1/2 = 1.50.

Q297: (A) $\alpha_m = 0.5 \ln((1 - 0.5)/0.5) = 0.5 \ln(1) = 0$. Instance weight update factors are $\exp(\pm 0) = 1$, leaving all weights unchanged.

Q298: 1 Multipliers are 10.0, 4.5, 10.0. The margin support vectors satisfy $0 < \alpha_i < C$ (where $C = 10.0$). Only $\alpha_2 = 4.5$ satisfies this strict inequality. Count = 1.

Q299: (A) Equal variances across classes cancel the quadratic terms x_j^2/σ^2 in the log-odds ratio, producing a linear decision boundary.

Q300: 4.00 At boundary $p = 0.5 \Rightarrow w_0 + w_1 x = 0 \Rightarrow -2.0 + 0.5 x = 0 \Rightarrow 0.5 x = 2.0 \Rightarrow x = 4.00$.

Q301: (A) The stable manifold theorem establishes that gradient descent with random initialization almost surely converges to local minimizers rather than strict saddles.

Q302: 6.00 Pre-activations at $x = 2.0$: $z_1 = 1(2) - 1 = 1 > 0 \Rightarrow a_1 = 1$. $z_2 = -1(2) + 1 = -1 \leq 0 \Rightarrow a_2 = 0$. $z_3 = 0.5(2) + 0 = 1 > 0 \Rightarrow a_3 = 1$. $z_4 = 2(2) - 5 = -1 \leq 0 \Rightarrow a_4 = 0$. Output $\hat{y} = 2(1) + 3(0) + 4(1) + 1(0) = 2 + 4 = 6.00$.

Q303: (A, B, C) FlashAttention, mixed-precision, and activation checkpointing optimize memory bandwidth and throughput.

Q304: 18 Vector $c_2 - c_1 = (4, 2)^T$. Midpoint $m = (3, 3)^T$. Hyperplane: $4(x_1 - 3) + 2(x_2 - 3) = 0 \Rightarrow 4x_1 + 2x_2 = 12 + 6 = 18$.

Q305: (A) The compact SVD provides an orthonormal basis for the column space of X via the first r left singular vectors U_r.

Q306: 0.231 Point 1: y=1, p=0.9 $\Rightarrow -\ln(0.9) = 0.1054$. Point 2: y=1, p=0.7 $\Rightarrow -\ln(0.7) = 0.3567$. Point 3: y=0, p=0.3 $\Rightarrow -\ln(1-0.3) = -\ln(0.7) = 0.3567$. Point 4: y=0, p=0.1 $\Rightarrow -\ln(1-0.1) = -\ln(0.9) = 0.1054$. Mean = (0.1054 + 0.3567 + 0.3567 + 0.1054) / 4 = 0.9242 / 4 ≈ 0.231.

Q307: (A, B, D) Points inside the tube incur zero loss and have $\alpha = 0$. Increasing ε widens the tube, decreasing the number of support vectors (C is false).

Q308: 0.60 Sum = 0.8(+1) + 0.6(−1) + 0.5(+1) + 0.4(−1) + 0.3(+1) = 0.8 − 0.6 + 0.5 − 0.4 + 0.3 = 0.60.

Q309: (A) Identical isotropic covariances reduce the Gaussian exponents to $-\|x - \mu_k\|^2 / (2\sigma^2)$, making assignments depend purely on squared Euclidean distances.

Q310: 40.0% Total variance = 12 + 6 + 2 = 20. First component explains 12 (60%). Lost variance = (6 + 2) / 20 = 8 / 20 = 40.0%.

Q311: (A, B, C) Adding an L2 ridge penalty λI guarantees invertibility by shifting all eigenvalues strictly positive, so (D) does not cause non-invertibility.

Q312: 3.61 $w_{new} = (1, -1)^T + 1 \times (2, 3)^T = (1 + 2, -1 + 3)^T = (3, 2)^T$. Norm = $\sqrt{(3^2 + 2^2)} = \sqrt{(9 + 4)} = \sqrt{13} \approx 3.61$.

Q313: (A) Early exaggeration inflates high-dimensional probabilities to form tight local cluster groupings before fine-tuning global placement.

Q314: 0.70 Sensitivity = TP / (TP + FN) = 30 / 50 = 0.60. Specificity = TN / (TN + FP) = 40 / 50 = 0.80. Balanced Accuracy = (0.60 + 0.80) / 2 = 0.70.

Q315: (A, B, C) ViT processes patch tokens globally through self-attention with minimal inductive bias, excelling primarily on large datasets.

Q316: 8.00 $\hat{\theta}_{UMVUE} = ((1 + 1)/1) \times 4.0 = 2 \times 4.0 = 8.00$.

Q317: (A) Smooth circular/spherical L2 balls touch quadratic loss contours tangentially off the axes; the sharp non-differentiable corners of L1 balls are what force sparsity.

Q318: 7.00 In Complete Linkage, $d(C_{12}, C_3) = \max(d(C_1, C_3), d(C_2, C_3)) = \max(4.0, 7.0) = 7.00$.

Q319: (A, B, D) GMM likelihood is non-convex with multiple local optima and potential singularities; global convergence cannot be guaranteed (C is false).

Q320: 8 Sum of eigenvalues equals the trace: Tr(S) = 4 + 4 = 8.

Q321: (A) The mini-batch estimate of mean and variance has sampling noise that injects stochastic perturbation into neuron activations, providing regularization.

Q322: 4 Pos: [0.9, 0.6, 0.4]. Neg: [0.7, 0.2]. 0.9 > 0.7 (yes), 0.9 > 0.2 (yes). 0.6 > 0.7 (no), 0.6 > 0.2 (yes). 0.4 > 0.7 (no), 0.4 > 0.2 (yes). Count = 2 + 1 + 1 = 4 pairs.

Q323: (A, B, C) Boosting sequentially fits residuals, which primarily reduces bias (D is false).

Q324: 0.67 Among the top 3 documents [Relevant, Non-Relevant, Relevant], 2 are relevant. P@3 = 2 / 3 ≈ 0.67.

Q325: (A) The double descent phenomenon demonstrates that heavily overparameterized models interpolate data while implicit regularization selects minimum-norm functions with smooth inductive bias.